Camosun College

Chem 150 Lab Manual

2021 Edition

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Lab 0 Safety in the Chemistry Lab

You must attend the safety lab before commencing with any other laboratory work. If you missed the safety lab you must make arrangements with the instructor for making up the safety lab prior to the next lab period.

The Chemistry Lab is an important component of Chem 150. You need to be familiar with all the safety features in the lab and know the procedures in case of an emergency:

1. The laboratory environment – what do I need to bring?
   As far as lab safety is concerned, the lab exists in two states: A lab is in progress or there is no lab in progress. We might meet in the lab for a review, to write a test or for office hours when there is no lab in progress. It is then permitted to enter Tec 230 without safety glasses, lab coat or other protective wear such as covered shoes. You are still entering a chemistry lab though and **at no time may you consume food or drink in the lab**.
   If there is a lab in progress you must wear your safety glasses the whole time you are in the lab. It is highly recommended that you also wear a lab coat for protection. In the absence of a lab coat you must wear a long-sleeved shirt or sweater, long pants and shoes that cover your toes. Safety glasses and lab coats might be purchased from the bookstore or any retailer (like Lumber World or Professional Uniforms). If you wear reading glasses you have the option of using googles that will fit over your glasses. Ask your instructor for them. There will also be disposable gloves available.

2. Spills:
   If a spill of any substance occurs in the lab, immediately notify the instructor.
   Spills on bench, floor or equipment:
   If a spill occurs on the lab bench or floor get the instructor's attention. It is important that a spill is cleaned up properly right away before it can cause harm or spread. A spill kit is located right behind the door to the chemistry prep room. Even water spilled on the floor constitutes a danger and must be sponged up immediately.
   If a spill of a substance occurs that involves your body disregard all modesty and take off the contaminated clothing. Wash your skin with plenty of water. If need be use the emergency shower. In the unlikely event of a substance entering your eye (You are wearing safety glasses all the time!), it must be flushed out with water. Use the eye wash fountain. Force the eye open and rinse for at least two minutes. Seek medical advice.
   Spills on an analytical balance:
   Spills on a balance should never occur when substances are transferred properly and care is taken. Here is how to use a balance properly. First: zero the balance. Second: place the empty weighing dish or crucible or beaker on the balance and record its mass. Third: remove the weighing dish/crucible/beaker from the balance and place it on the clean lab bench beside the balance. Fourth: transfer the substance using a spatula. Fifth: carefully place your loaded weighing dish on the balance and record the combined mass of. **Substances must NEVER be transferred over the balance**. Also: **Never weigh a hot object!** Give hot crucibles plenty of time to return to ambient temperature before weighing.

3. Broken Glass:
   Handle glassware with care. If you discover broken glassware in your drawer notify you instructor. Do not carry out an experiment with cracked or chipped glassware. If you break glass, ask the instructor for broom and dustpan and clean it up immediately. Broken glass constitutes a serious hazard. **Broken laboratory glass is disposed of in a special container. Do not place it in the garbage or the recycling**.

4. Fire
   If the fire bell sounds stop what you are doing. Make your experiment safe (e.g. turn of the Bunsen Burner or turn off the hotplate). We will leave the lab and the building and walk toward Campus Centre and assemble on the grass to the left of the bottom of the stairs that lead to the building. This is next to the bike barn.
   If we are the cause of the fire, e.g. there is a fire in the chem lab, assess the situation: Is this a small fire that can be dealt with or is this beyond our means of fighting a fire. For a small fire (for example 100 ml of ethanol are spilled on a lab bench right next to a Bunsen burner and caught fire) use one of the three fire blankets to smother the flames. There is also a fire extinguisher located near the front desk by the printer. If the fire is serious, we leave the lab and pull the fire alarm located about 20 m left down the hallway leaving Tec 230. Leave all your personal belongings behind. Just try to make your experiment safe. The main gas valve should be shut off. Assemble in the same area as mentioned above. The instructor will do a dry run of an evacuation so everyone knows where to assemble.

5. Earth quake
   In case of an earth quake hold on and cover. The lab benches provide good cover but use good judgement if they are in use with a procedure such as boiling large quantities of water. After the shaking has stopped make your experiment safe and follow the evacuation procedure described above under 'fire'. Do not return to the building unless you have been assured that it is safe to do so by an authority such as a warden.

6. Electrical
   Every lab bench is equipped with four electrical outlets. If more than two hotplates are plugged in on the same circuit the breaker switch might be coming off. The instructor has no means of reinstating the power. Facilities have a key to the breaker box and will need to be called. This could take a substantial amount of our lab time so please be careful not to overload the circuit. In case of an electrical emergency the power can be shut off using the emergency button next to the door (with the Plexiglas guard around it and the sign "This is not a light switch"). Several of the labs will involve the use of laptops and data loggers next to hoses with running water or hotplates with beakers containing electrolyte solutions. Be extremely careful not to spill water or solutions on the electrical equipment.

7. Clean up
   Keep your lab space clean and tidy. Spills tend to happen more frequently in a cluttered environment. If you are finished with an experiment, clean all equipment and return it to its proper location.

8. Judgement and Care
   You will be working with acids, bases, heavy metals and other potentially harmful substances in the lab. Use caution and good judgement when working with those hazardous materials. Inform yourself ahead of the lab about the hazards and dangers of the substances you will be working with. Dispose of chemicals properly. There will be waste bins or beakers in the fume hood. Wash your hands before you leave the lab. Use the fume hoods when instructed.

Lab 1 Densities of Liquids and Solids.

Purpose

The purpose of this introductory lab is to identify an element from a set of possible candidates by measuring its density. This lab will familiarize you with analytical balances and the use of some volumetric glassware. This lab also serves as an exercise in reporting experimental results with the correct number of significant figures.

Theory

Density is a distinct property of matter. Because it is a ratio, the ration of mass divided by volume of a substance, is independent of the amount of substance. Such a property is called an intensive property. Other examples of intensive properties include boiling point, cell potential or vapour pressure. In contrast, an extensive property does depend on the amount of substance e.g. the amount of heat (enthalpy) produced in a combustion reaction will depend on the amount of fuel combusted.
The density of a substance will depend on the temperature and to a lesser extend on the pressure. In general, densities are determined at standard temperature and pressures. For liquids, solids and most purposes in the laboratory standard temperature refers to 25°C (298K) and 1 atmosphere (101.325kPa) (i.e. room temperature and average atmospheric pressure at sea-level).
The precision of an experimentally determined density will depend on the precision of all measurements and the accuracy will depend on precision and on the purity of the substance being measured.
In this exercise, the quality of the result will depend partly on your skills and partly on the quality of the instruments and materials.

If you are curious about the density of an object you might want to bring it to this lab and we will try to measure it. (i.e. is a precious ring really made of pure gold, what is the density of a mineral…). The sample should fit in a small vial with a 2 cm opening.

Procedure

Part 1: Solid

Determine the precise mass of a clean dry container (including lid) on the **analytical balance** to 1 mg. Add a few pieces of the selected metal (for most samples about half full, for dense samples about one third) and determine the combined mass. Next fill the container with the metal with deionized water (dH_2O) and determine the mass of container, solis, water and lid. When filling the container, add water and ensure that no air is trapped on the metal bits. You might want to carefully stir with a glass rod to eliminate attached air bubbles. Continue to add water until there is a positive meniscus showing above the lip. Slide the lid over the meniscus and lower into place, ensuring that no air is trapped. Dry the outside of the weighing dish of all spilled water before weighing. Empty out the water from the weighing dish taking care that non of the sample is lost and return the unknown sample to the drying dish in the fume hood with the same number as your unknown sample. Please do not mix the unknown samples. Last fill the container without any sample with dH_2O in the same way, ensuring no air entrapment. Repeat this measurement once e.g. empty the water from the container and refill, add the lid, dry and reweigh to get a sense of how reproducible your results are.
These data permit you to calculate the density of the solid. Which of the unknown metals do you have? Note that on the table below, some of the possible metals have very close (or the same) values. You will have to provide either/or.

Part 2: Liquid

Precisely determine the mass of a clean dry pycnometer to 1 mg. Fill the pycnometer with d H2O and determine the precise mass again. Empty the pycnometer, rinse twice with unknown, and fill with unknown. Determine this precise mass and calculate the density of the unknown.

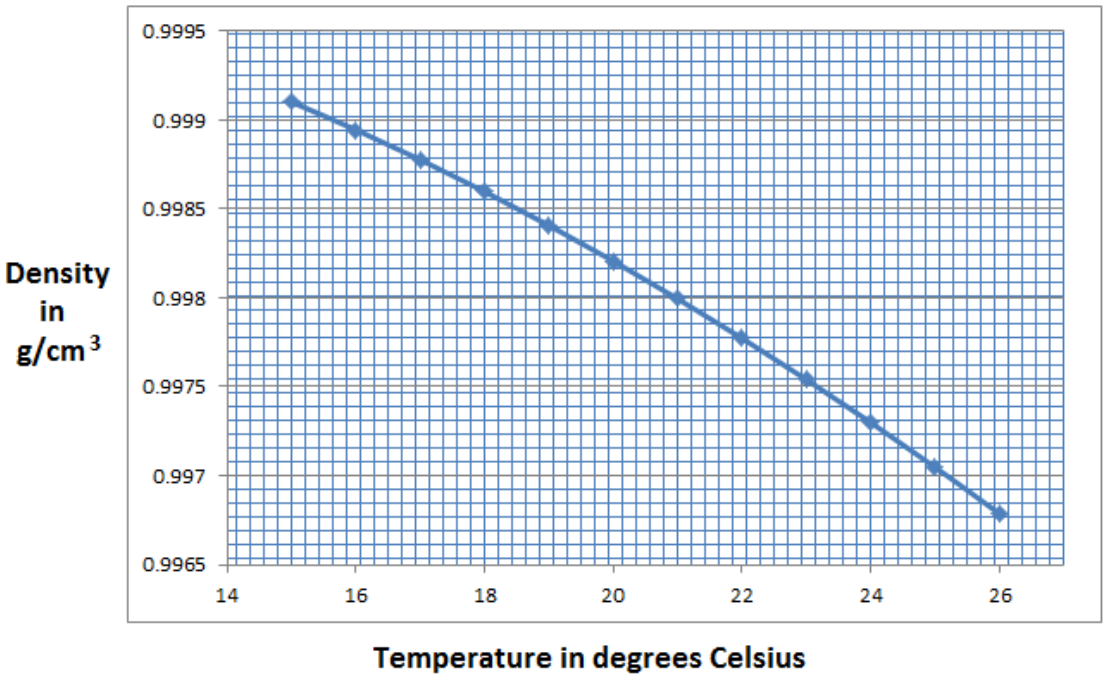
You are all measuring the same unknown liquid. Therefore, you should all get a similar density but do not expect to get exactly the same value as your colleagues, because all measurements have inherent imprecision.

Data analysis and report

Enter all raw data and your final calculated densities on the data sheet. You can show calculations on the back of the data sheet. Be sure your final answer includes the appropriate number of significant figures.

Metal	Density(g/cm^3)	Metal	Density(g/cm^3)	Metal	Density(g/cm^3)
Na	0.97	Ce	6.76	Mo	10.2
I	1.49	Zn	7.14	Ag	10.5
Mg	1.74	Sn	7.31	Pb	11.4
Si	2.33	Fe	7.86	Pd	12.0
Al	2.70	Ni	8.90	Hg	13.6
Ti	4.51	Co	8.90	Au	19.3
Ge	5.32	Cu	8.90	Os	22.4

Density of water as a function of Temperature at a pressure of one atmosphere



Source: Handbook of Chemistry and Physics 90th Edition 2009-2010

Data sheet Laboratory #1: Densities of solids and liquids

Temperature in the laboratory _____

Density of water at that temperature_____

Part 1: Solid

Unknown Solid # _____	1th measurement	2nd measurement	average
Mass of empty dry flask with lid			
Mass of flask with solid and lid			
Mass of flask with solid, water and lid			
Mass of flask full of water with lid: 1st try			
Mass of flask full of water with lid: 2nd try			
Average of 1st and 2nd measurement			
Volume of solid			
Density of solid			
What was the solid			

Part 2: Liquid

Mass of empty dry pycnometer	
Mass of pycnometer with water	
Mass of pycnometer with unknown	
Volume of pycnometer (give all significant figures!)	
Density of unknown liquid	

Lab 2 Stoichiometry

Stoichiometry refers to the ratio in which elements combine to form compounds or the proportions in which compounds themselves react to give products. Atoms reside in molecules as whole number proportions – one does not find parts of atoms in a molecule, because atoms are the smallest unit of an element. So, the Law of Definite Proportions states that elements combine in ratios of small whole numbers.

In this experiment you get to investigate the law of Definite Proportions by carrying out a synthesis of a compound from two elements (Case 1) and by conducting a decomposition of a compound into simpler substances (Case 2). By carefully weighing reactants and products you will be able to determine the ratio of atoms in either case.

Case 1: sulfur reacts at high temperature with copper to form a compound Cu_xS, where ***x*** is an unknown number. You will undertake this reaction and determine the value of x. The reaction is:

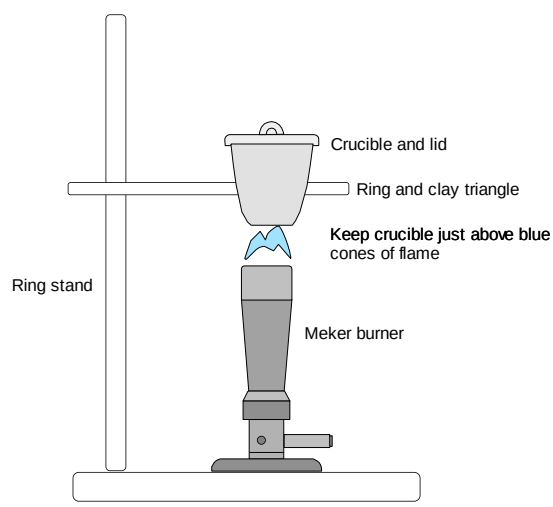
$$\boldsymbol{x}Cu + S \rightarrow Cu_{\boldsymbol{x}}S$$

Case 2: you have a compound $CuBr_x$, where ***x*** is unknown. **Gentle** heating of this compound gives off bromine gas, Br_2, and leaves behind CuBr, i.e., the compound with one atom of copper and one atom of bromine. The reaction is:

$$CuBr_{\boldsymbol{x}} \rightarrow Br_2 + CuBr \qquad \text{(Not balanced!)}$$

Case 1: Copper and sulfur.

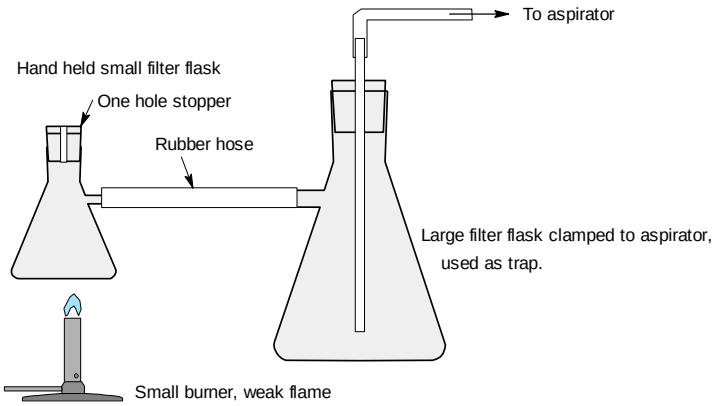
Perform this experiment in the ***fume hood (turned ON)***.



The instructor will demonstrate the apparatus. First, heat the empty crucible and lid until red hot, allow to cool*, and then determine the mass of the empty crucible on an analytical balance. Prepare a **1 g** coil of copper from the wire provided, place it in the crucible, and determine the precise mass of copper and crucible. Use calipers to measure the diameter of the wire. Add **about 1.2 g** sulfur (precision not required) and heat the crucible to red heat until no vapours or flames appear around the lid. Heat for another two minutes and let the apparatus cool. Determine the precise mass. Add **~0.5 g** sulfur, heat again, cool and determine the precise mass once again. If the masses of both weightings agree within 5 mg move on to the calculations otherwise repeat this procedure until the mass of the flask and ***product does not change (within 5 mg) or decreases.*** These data permit you to determine the value of ***x*** in the reaction equation. See the data sheet provided. *__Never weigh a hot object!__

Case 2: Copper bromide

This exercise will be done on the bench, using a vacuum trap system to capture the noxious fumes of bromine. The instructor will demonstrate the apparatus.



Determine the precise mass of a clean, **dry** small filter flask. Add **1 g** of copper bromide provided, and determine the precise mass again.

Set up the apparatus as above, and heat the copper bromide ***GENTLY*** while applying suction to the system. After all of the copper bromide has changed to a pale green colour, cool, and then determine the precise mass of the flask and product. Repeat this procedure until the mass of the flask and ***product does not change (within 5 mg) or increases***. If your product turns dark again, you have heated too strongly or for too long. These data permit you to determine the value of ***x*** in the reaction equation.

Cleanup

Try to dump out as much of the CuBr as possible into the provided container before washing the filter flask with soap and water and place them in the oven to dry overnight. The crucibles and lids should not be washed. Simply dump the product in to the container provided labeled Cu_xS Don't forget to measure the diameter of your product first!.

Data analysis and report

Collect your data directly into the data sheet. Calculate all quantities required in the data sheet. Give sample calculations on the back. Attach more if necessarily. Use appropriate formulas to do the calculations, carry all units and give your final results with appropriate number of significant figures.

Data sheet Laboratory #2: Stoichiometry

1. Copper and Sulfur

Diameter of copper wire:	
Mass of empty crucible:	
Mass of crucible plus copper:	
Mass of crucible and product after 1st heating:	
Mass of crucible and product after 2nd heating:	
Mass of crucible and product after 3rd heating (if req.):	
Diameter of product "wire"	
Mass of copper:	
Mass of product:	
Mass of sulfur that combined with the copper:	
Ratio of moles of copper to moles of sulfur (all sig.figs!):	
Empirical formula for Cu_xS (one sig fig):	
Volume of copper using ($density_{Cu}$ = 8.90 g/cm^3):	
Volume of product:	
Density of product:	

2. Decomposition of $CuBr_x$

Mass of empty flask:	
Mass of flask plus copper bromide:	
Mass of flask plus decomposed copper bromide 1st weighing:	
Mass of flask plus decomposed copper bromide 2nd weighing:	
Mass of flask plus decomposed copper bromide 3rd weighing (if req.):	
Mass of decomposed copper bromide:	
Number of moles of decomposed copper(I)bromide (n CuBr):	
Number of moles of bromine gas given off (n Br_2):	
Ratio of moles of Cu to moles of Br in $CuBr_x$ (give all sig. figs) ($\frac{n\ Cu}{n\ Br}$)	
Empirical formula for $CuBr_x$ (one sig. fig):	

Lab 3: Spectroscopic Determination of Nickel

Background (We will cover some of this in class in Quantum Mechanics.)

Much of the understanding of the modern structure of the atom came from experiments that revealed how light interacts with matter. Electrons in atoms (or molecules) can absorb energy and might be excited to energy levels higher than the ground state by flame or high voltage. They can also be excited by light. When electrons return to the energetically lower (more stable) ground state the excess energy is given off and the atom emits light giving rise to emission spectra. While emission spectra arise from transitions from an excited state to a lower state, absorption spectra arise from transitions from a lower energy state to a higher energy state.

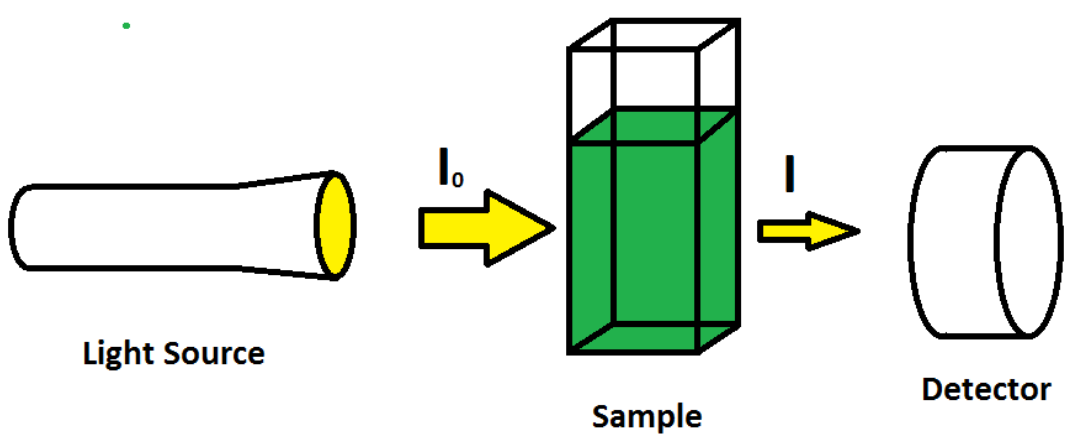
In the gas phase, atoms are isolated and transitions are usually very sharp with well defined transition energies. In solution, however, many processes contribute to energy changes, because the atoms or ions are in environments which modify the basic energy levels. The result of this is that the sharp line spectrum is modified to give a broad absorption band.

Where a line spectrum would be ideal for the analysis of a metal or ion, the broad absorption found in solutions is not ideal, and the best part of the absorption curve must be found to give the most sensitive analysis. Usually, the best position on the curve is the peak, and if the peak is broad, small wavelength variations do not cause a problem.

In this exercise, you will determine the best wavelength for the nickel analysis, and then construct a calibration curve using standards with a known nickel concentration. You will then determine the concentration of an unknown nickel solution.

Solutions of nickel are green because they absorb the complimentary colour from a white light source. Imagine a beam of light of intensity I_o entering a sample. The sample will absorb some light and the light beam exiting the sample will have a diminished intensity I. Absorbance is defined as the log of the ratio of the intensity of those light beams. Absorbance = log $\frac{I_0}{I}$

It is thus a dimensionless number. When $I = I_0$ the ratio is 1 and log 1 =0, the absorbance is zero and all the light is transmitted. If 99 % of the light is absorbed and only 1 % is transmitted the absorbance is log $\frac{100}{1}$ = 2.



In general, the absorbance of light by a solution obeys Beer's Law, which is written

$$\mathbf{A = \varepsilon\, b\, c,}$$

Where **A** is the absorbance, **ε** is the molar extinction coefficient, **b** is the length of the path the light traveled through the sample, and **c** is the concentration. For a given set-up **b** will be constant and **ε** will be constant for a particular wavelength and temperature, so **A** will be proportional to **c**. It is important to use the same cuvette for all measurements, and to place the cuvette in the sample holder with the same orientation each time and measure all standards and the unknown at the same wavelength.

Part 1: Preparing a set of calibration standards using 50 mL volumetric flasks

You are provided with a nickel solution of known concentration. Record the molarity. Set up a 50 mL burette and charge it with the standard nickel solution. If the burette is wet it must be rinsed with stock solution first. Collect all nickel waste solutions. Do not dump any of it down the drain.

Obtain four flasks and label them with the codes 1, 2, 3 and 4. To these four volumetric flasks add approximately 37.5 mL, 25 mL, 12.5 mL and 6.25 mL of nickel stock solution from the burette. It is not necessary to add exactly those amounts, but it is necessary to know exactly how much you added to each (see data table). After adding the standard nickel solution, make up the volume in each flask to the mark with D_{H_2O}. Place a lid on each flask and mix thoroughly by inverting the flasks 10 times.

Part 2: Wavelength determination

The instructor will demonstrate the use of the Spectronic 20 visible spectrometer. This instrument consists of a light source followed by a monochromator, which disperses the spectrum of light from the source. A wavelength dial selects the wavelength of light which passes through a slit then through the sample and onto a photomultiplier tube. The photomultiplier tube output depends on how much light passes through the sample, so samples that absorb the light will result in lower output.

The Spectronic 20 is a single beam instrument, which means that only one light path is used. As the energy output of the lamp varies with wavelength, the instrument **must be calibrated each time the wavelength is changed**.

With no sample in the sample chamber, set the meter to ∞ absorbance using the left knob. This sets the current of the photomultiplier to zero and will not be adjusted again although it will be checked after all other measurements are made.

Set up the instrument by adjusting the wavelength dial to scale, using the neat standard (highest concentration as provided). Adjust the wavelength to 370 nm. Fill one cuvette with D_{H_2O} and one with stock standard. Place the cuvette with D_{H_2O} in the sample chamber and set the meter to A=0 using the right knob. Next place the cuvette with standard in the sample chamber and record the absorbance. Ensure the lid to the sample compartment is closed each time you take a reading or zero the instrument to avoid ambient light from reaching the detector.

Readjust the wavelength to 380 nm and repeat the calibration: measure D_{H2O} followed by sample, change λ, measure D_{H_2O}, measure sample, change λ, etc. Record the absorbance at each wavelength.

Determine the maximum absorbance, and set the wavelength to that value.

Part 3: Preparing the unknown

Obtain an unknown sample from the instructor and **record the unknown number**. **Each student will do an unknown**. Take a clean 50 ml volumetric flask and a clean funnel (both need to be clean but not dry). Transfer all unknown solution in the 50 ml flask. Rinse the vial that contained your unknown at least two times with a few ml of water to ensure all the nickel is transferred to the volumetric flask. Rinse all unknown solution from the funnel (careful, do not overfill the flask) and finally make up to the mark with water. Stopper the flask and mix by inverting the flask 10 times. Label your unknown!

Part 4: Calibration and measuring the unknown

At the wavelength of maximum absorbance, determine the absorbance for the four calibration standards you prepared and the unknown. Start by calibrating with water to set zero absorbance. This is your first data point in the calibration. Empty the water from the cuvette and rinse with a small amount of your most dilute standard. Rinse once more and fill the cuvette. Measure your first calibration standard. Repeat for all standards and your unknown. Rinse each time you change solutions in the cuvette to avoid diluting the standards. At last measure undiluted stock solution. With the D_{H_2O} and the stock solution, you now have six calibration points.

Data analysis and report

Prepare a plot of absorbance (y) *vs.* wavelength (x) to demonstrate the shape of the absorbance curve. Label both axis and give your graph a meaningful title. Hint: *Absorbance vs. Wavelength* is not a very useful title!

Next plot the absorbance of the calibration standards *vs.* Molarity of Nickel and find the line of best fit (linear regression). Your graph should have six data points: the four standards, undiluted stock solution and pure water. Make sure you include the zero measurement as a data point (e.g. when you calibrated your instrument against pure water the molarity of nickel in pure water was 0 M and the absorbance was zero). Do not force the line of best fit through the origin (excel will have that option). Show the equation of the line of best fit and the R^2 value on the graph. Rearrange the equation so obtained and solve for molarity. Using the absorbance of the unknown, calculate the unknown concentration and convert the result to mg Nickel in the unknown sample.

Data sheet: Spectroscopic Determination of Nickel

Part 1: Preparing calibration standards and Part 4, calibration.

Stock nickel (Ni^{+2}) concentration: _____ M

	Part 1		Part 4
#	**Volume stock added to 50 mL**	**Calculated concentration (M)**	**Measured absorbance at _____ nm**
0	0.000 mL	0.000 M	
1	mL	M	
2	mL	M	
3	mL	M	
4	mL	M	
5	Undiluted stock	M	

Part 2: Wavelength determination

λ (nm)	A	λ (nm)	A	λ (nm)	A	λ (nm)	A
370		430		490		550	
380		440		500		560	
390		450		510		570	
400		460		520		580	
410		470		530		590	
420		480		540		600	

Part 4: Measuring the Unknown.

Student:	Unknown #	Unknown Absorbance	Molarity	Mass Ni (mg)
Name_____				
Name_____				

Lab 4 Shape of Molecules and Polarity

In this course we have already made predictions of the physical world based on an understanding of structure and bonding on an atomic or molecular level. Remember the trends in melting points, solubility and lattice energy of simple ionic compounds and our rationalization of these trends combining concepts of bonding, Coulomb's Law and trends in atomic properties. This structure-property relationship is indeed one of the most powerful tools of chemistry and its theories.

A very important question we need to be able to answer is that of polarity. Polar molecules behave completely different compared to non-polar molecules. Many physical properties such as vapour pressure, viscosity, melting points and solubility depend on the type and strength of forces acting between molecules. These forces are to a large degree influenced by polarity.

To decide if a molecule is polar or not we need to know the three dimensional shape of the molecules. If there are multiple possible Lewis structures that can be drawn for a molecule we also need to consider the resonance hybrid.

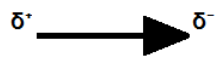
<u>The origin of polarity</u>:

In homonuclear diatomic molecules like H_2, O_2 or N_2 electrons are shared equally. The resulting bond is said to be non-polar. All heteronuclear diatomic molecules like HCl or BrI must be polar because the bonding electron pair is attracted more strongly to the element with the greater electronegativity. This results in partial charges on the elements which we indicate with a δ^- and a δ^+on the atoms.

Thus in HCl hydrogen the carries a δ^+ and chlorine, the more electronegative element carries a δ^-. The magnitude of the partial charge varies depending on the difference in the electronegativity (ΔEN) of the atoms. The greater ΔEN, the more polar the bond. In HCl (g) measurements indicate that the magnitude of the charge on each atom is about 17 % of that of an electron. It is really only a partial charge, a fraction of a formal elemental charge. This imbalance of charge creates a bond dipole.

Polar bond resulting in a polar molecule

polar bond reduced to "antenna" with atoms representes as point masses at each end

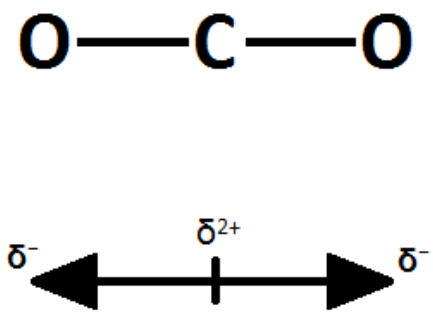


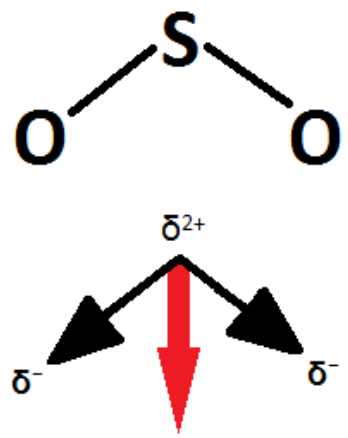
A bond dipole is a vector. It has derection and magnitude.

Dipoles have vector properties and if a molecule has several bonds (e.g. any molecule made up of three or more atoms) the individual *bond dipoles* must be added up so see if the overall molecule is polar or not. The overall *dipole moment* μ is also called the *resulting dipole moment* of the molecule.

If we consider larger molecules with multiple bonds it is necessarily to know the three-dimensional shape of the molecule to understand how the individual bond dipoles add up. In very symmetrical molecules they might in fact cancel out completely and we can have a molecule with polar bonds, even an ion with a charge that is non-polar. Consider CO_2 and SO_2.

CO_2 is a linear molecule and the two bond dipoles counteract and cancel each other out. CO_2 is non-polar. SO_2 on the other hand is bent and the two bond dipoles add up to give a resultant dipole moment (red arrow). SO_2 is polar!





The shape of molecules:

To deduce the shape of a molecule we use Valance Shell Electron Pair Repulsion theory (VSEPR-Theory) Start by drawing the Lewis structure of the molecule or ion in question. You don't have to draw the best Lewis structure but you have to draw a valid Lewis structure. Count the number of electron groups (#EG) around the central atom. Bonding electron pairs, lone pairs or multiple bonds count as one electron group. Next define the electron group geometry (EGG). Electron groups repel another. They orient themselves as far apart as possible to minimize repulsion:

# EG	EGG
2	linear
3	trigonal planar
4	tetrahedral
5	trigonal bipyramidal
6	octahedral

After defining the EGG draw it. If there are lone pairs place them first and continue to place bonding electron pairs around the central atom. Recall that in trigonal bipyramidal EGG lone pairs are always placed in the equatorial positions. Now draw the shape of the molecule. Make sure it looks three-dimensional. Use the solid wedge for electron groups coming out of the plane of the paper and the broken wedge for EG's that point into the plane of the paper. Finally, state if the species is polar or not.

Part A **Basic Shapes**

Complete the table below by drawing Lewis structures. Identify and draw the EGG and the molecular shape of the species in question. Make a model for each of the molecules and draw it as you see it. Finally decide if the molecule or ion is polar or not.

Species	Lewis Structure	#EG	EGG	shape (Draw it as you see it!)	polar?
CO_3^{2-}					
NH_4^+					
NH_3					
PF_5					

SF_4

SF_6

BrF_5

Part B **Molecules with more than one central atom**

Make models of Acetone $(CH_3)_2C{=}O$ and hexane C_6H_{14}. **Keep the hexane model for part D!**

What is the EGG of every C-atom in hexane: _____

What is the EGG of the C-atom in acetone bound to oxygen: _____

What is the EGG of the C-atoms in acetone bound to hydrogen: _____

Is acetone polar or non-polar:_____

Is hexane polar or non-polar:_____

Part C **Visualizing Polarity**

Proceed to the front of the lab where some burettes have been set up.

Produce static electricity by rubbing a glass rod or Teflon rod against your lab coat or a cloth. Let your lab partner open up the burette containing water producing a small stream of liquid running out in to a catchment beaker. Bring the charged rod near the tip of the burette without touching it or the liquid.

What do you observe? Record your observations for water:

Repeat for Acetone and record your observations:

Repeat for hexane and record your observations:

Part D **Conformers**

You might have noticed that the static electricity produced a tiny effect on the hexane in Part C. But how can this be if hexane is completely non-polar? The reason for this is the presence of many conformers of hexane. There is free rotation about every C-C single bond in hexane and at room temperature many different conformers are present in the liquid which are quickly interchanging.

Use your model of hexane you made in Part B. Arrange the chain in a zigzag fashion so all carbons are as far away from each other as possible. This is the non-polar conformer of hexane. Next twist and rotate the chain and produce a less symmetrical conformer. Sketch this conformer.

Is it completely non-polar? Explain!

Conformers of cyclohexane C_6H_{12}.

Make a model of cyclohexane C_6H_{12} by slightly modifying your hexane model. The conformation which has each carbon atom as far away as possible from each other is called the chair-conformation. Sketch this conformer. Is it polar?

Next bend the ring into the boat-conformation. Sketch the conformer. Is it completely non-polar?

Part F **Isomers**

Structural (constitutional) isomers:

There are two different ways the molecule $\mathbf{C_2H_6O}$ can be arranged. Make models of both of them and give Lewis structures below.

Structural Isomer A Structural Isomer B

Isomers-unlike conformers do not easily interchange since chemical bonds need to actually break and reform (not just rotate).

Stereoisomers also known as Geometric Isomers:

These are molecules that have the same constitution but differ in the three-dimensional orientation of the atoms in space. The molecule 1,2-dichloro ethene, H(Cl)C=C(Cl)H exists in two geometric isomers. The C=C double bond prevents a rotation about the two carbon atoms making the molecule ridged. Build molecular models of both isomers and sketch them below. State if each isomer is polar or not. The isomers are referred to as *cis* and *trans* isomers of 1,2-dichloro ethane. Both isomers have different physical and chemical properties.

Geometrical Isomer A

Geometrical Isomer B

Another form of Stereoisomers are Enantiomers. Enantiomers (optical isomers) are non-super imposable mirror images of two molecules. They behave like right and left hand. A left glove will not fit a right hand. Such molecules are said to be chiral. They have the same physical properties except that they rotate the plane of polarized light in opposite direction (by the same amount). They also differ in one more important aspect, their reactivity towards other chiral molecules.

Make a model of chloro bromo iodo methane. Sketch the molecule. Next prepare a model of the other enantiomer. Ensure that the two are mirror images of each other that cannot be made to coincide with each other.

chloro bromo iodo methane *enantiomer1*

chloro bromo iodo methane *enantiomer 2*

Lab 5: Distillation

Introduction

Distillation is a process by which a volatile liquid is heated to evaporation, passed through a condenser which returns it to liquid state, and is collected. If the liquid is part of a mixture, the liquid can be separated from the mixture if the boiling point of the liquid is different from the boiling points of the other components of the mixture. One common example of distillation is the purification of water from an impure source, such as waste water (rare) or sea water (first described by Aristotle but now common in water deficient countries with low energy costs). Another is the refining of crude oil to produce various hydrocarbon fractions such as gasoline, diesel, kerosene, and bunker fuel (used by the Romans to produce turpentine from pine oil) or to purify silicon for the electronic industry by fractional distillation of $HSiCl_3$.

Another example is the distillation of alcohol spirits from fermented product, such as whisky (malt), rum (sugar or molasses), and vodka (wheat grain). Brandy and schnapps are produced from wines or fruit ferment. Sometime around the sixth century AD, Arabs, possibly Egyptians, possibly Armenians, made use of distillation to improve the quality of alcoholic beverages. One problem with fermented spirits is that methanol (CH_3OH, bp 64.7°@760 torr, and a toxic alcohol) is produced as a byproduct. Because methanol boils at a lower temperature than ethanol ($CH_3\ CH_2OH$, bp 78.5°@760 torr), discarding the initial output and stopping the collection when the temperature increased resulted in better and safer product.

Ethanol boils at 78.5ºC, while water boils at 100ºC. In fact, ethanol forms an **azeotrope** with water, at 95.6% ethanol/4.4% water, bp 78.2°@760 torr If the temperature of the mixture can be held above 78ºC and below 100ºC, then a separation will occur. The evaporated material is also a mixture of the two components, as both are volatile. In practice, this is a mixture of compounds depending on the respective vapour pressures at the temperature. A simple distillation will result in a new mixture with a higher fraction of the lower boiling material, but not a pure product. Permitting the evaporated material to reach some equilibrium temperature as it passes to the condenser can improve the efficiency of the separation, in effect by having a theoretical series of simple distillations occur en route.

In this exercise, you will compare a simple distillation of a cheap red wine to a fractional distillation of the product of the first distillation. The fractional distillation will start with considerable advantage, as the feedstock will have already been concentrated in ethanol by the initial simple distillation.

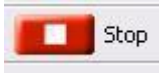
Method

Prepare two evaporating dishes (ceramic). When material is available, put a small amount of product from the simple distillation, and a small amount of product from the fractional distillation on a glass, and attempt to ignite the liquid with a burning splint. Keeping 5 ml of the product of the simple distillation should suffice for the flammability test. Use the bulk of the product to carry out the fractional distillation.

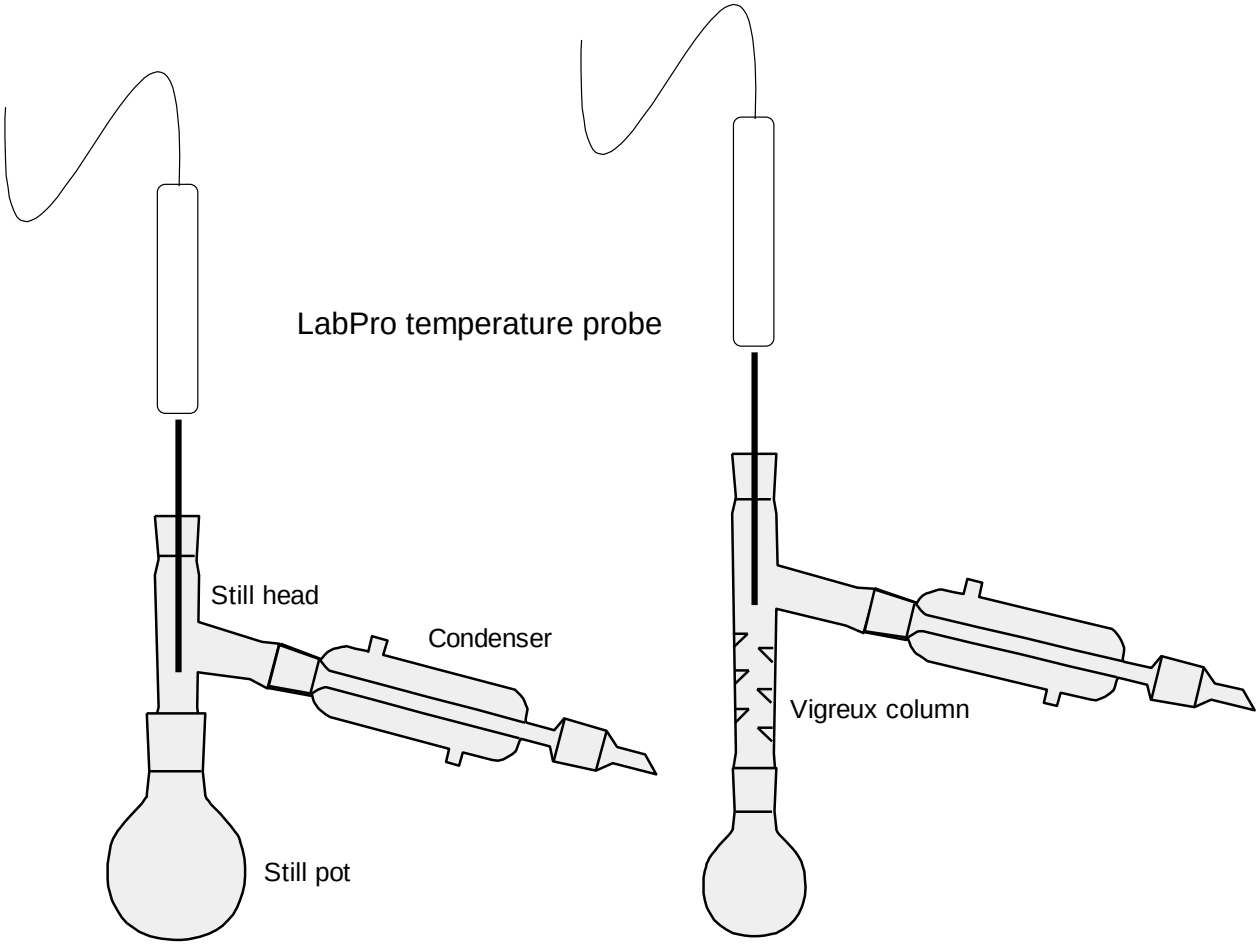
Set up the apparatus according to the demonstration by the instructor. Charge a 500 mL round-bottom flask with 250 mL of wine provided, add boiling chips (1 or 2).

Set up the data collection to the ***Events with entry*** mode, with the entry column labeled ***Volume*** and the units ***mL***.

Apply heat carefully with a ***micro*** Bunsen burner. Apply heat to keep the charge boiling ***gently***, not vigorously, so that the vapour just passes over to the condenser. Start the data collection. When the first drop of condensate drops into your collection vessel, press the ***Keep*** button

and enter 0 as the volume. Collect the condensate and record (***Keep***) the temperature at the head of the condenser for each 5 mL of condensate. Collect until the temperature at the condenser head reaches 98ºC or you collected 40 mL of distillate (whatever comes first). Stop the data collection 

and remove the Bunsen burner.



Simple distillation **Fractional distillation**

Set up the apparatus again including the Vigreux column as shown by the instructor. The Vigreux column permits the vapour to equilibrate at various temperatures as it rises, enabling better separation of the two volatile liquids. Charge a small round-bottom flask with most of the product of the simple distillation, add boiling chips, and apply heat. Save the previous run in LoggerPro, and again, collect the condensate and record the temperature in the same way. Collect data as above, except every **2 mL**. Collect until the temperature **starts to rise sharply**.

Data analysis and report

The object of this exercise is to compare simple to fractional distillation and make a comment on relative efficiency. Support your claim with data and observations from this lab (flame test, temperature profile for both distillation). If this were an experiment, one would properly conduct each distillation with the same feedstock, but experience has shown that although an improvement is seen, good separation does not occur, as it should with successive simple then fractional distillation.

You should have both the simple and fractional distillation in the same LoggerPro graph. If you prefer working with Excel you can copy the data columns into Excel and create your own graph. Use ***Insert Text Annotation*** to label the two curves. In the ***Graph options*** window, select ***Manual*** under ***Scaling***, and adjust the Y axis so that only the relevant data show.

Add a ***Text annotation*** to the graph, and briefly indicate the difference in burning of the two products, with a very brief explanation.

Time permitting, we will determine the density of your final product from the fractional distillation using the procedures for the densities of liquids from Lab 1. This will permit us to quantify the % vol ethanol content in your sample.

Lab 6 Recycling Copper

The purpose of this lab is to familiarize yourself with some fundamental terms in chemistry such as oxidation-, reduction-, precipitation- neutralization- and decomposition reactions. You will take elemental copper through a series of chemical transformations and eventually recover the copper metal. The mass of the recycled copper will be compared to the initial mass of copper at the beginning of the lab.

Procedure:

<u>Part 1 Conversion of copper to copper nitrate</u>

a) Measure out about 1.5 g of copper. Record its mass to the nearest 0.01 g

Starting mass of Cu(s) _____

b) Place the copper in a 250-mL beaker. Label the beaker. Using the dispenser in the fume hood add the specified volume of half concentrated nitric acid. (Caution: Use extreme care in handling of this acid. Wipe off any spills immediately even if it's only a drop! Nitric acid reacts with protein such as your skin, so ware gloves.)

c) Place a watch glass over the beaker. Set the beaker aside in the fume hood and record your observations. Be explicit!

_____

_____

_____

_____

_____

Copper is reacting with the nitric acid to form copper(II)nitrate, nitrogen monoxide and water. Oxygen from the surrounding air spontaneously reacts with NO(g) to form the brown NO_2(g). Balance this redox-reaction (**red**uction-**ox**idation-reaction):

$$Cu(s) + HNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + NO(g) + H_2O\ (l)$$

Part 2 Conversion of copper nitrate to copper hydroxide

a) When all the copper has dissolved, carefully take your beaker and rinse off any drippings, that might cling onto the watch glass *into* the beaker. Place a strip of blue litmus paper on a clean watch glass and determine the acidity/basicity of your reaction mixture using a glass rod to transfer just one drop of the solution.

Result: _____

There is un-reacted nitric acid left over which has to be neutralized.

b) Take 15 - 20 mL of 6M sodium hydroxide solution (caution: very strong base). Test one drop of this solution with red litmus paper.

Result:_____

c) Add a walnut size chunk of ice to your beaker. Stir as you **slowly** add the 20 ml of NaOH solution to the copper. Note what happens to the ice. Test the resulting solution with red litmus paper. If you don't get an alkaline reaction add more NaOH solution until you do.

Observations: Heat produced? (Ice melted?) _____

Litmus test:_____

Appearance of the mixture in the beaker:_____

Two chemical reactions have occurred. First the sodium hydroxide solution neutralized the excess nitric acid that was still present from part 1. Write a balanced equation for this acid-base reaction:

_____

The other reaction involves the formation of copper(II)hydroxide which precipitates out of solution due to its very low solubility in water. Write a balanced equation for this precipitation reaction:

_____

Part 3 Decomposition of copper hydroxide

a) Add about 100 mL water to the beaker containing the mixture from part 2. Heat on a hotplate to a **simmering** while stirring. *Careful! Do not boil! This heterogeneous mixture is prone to overheating and may suddenly boil with heavy splashes. Avoid overheating. Turn heat right down as soon as mixture has started to simmer and constantly stir to avoid bumping.* Continue stirring the hot mixture until the colour change is complete. Carefully remove the beaker from the hotplate and remove the stirring rod from the beaker rinsing off any solids. Allow the contents of the beaker to settle for 5 minutes.

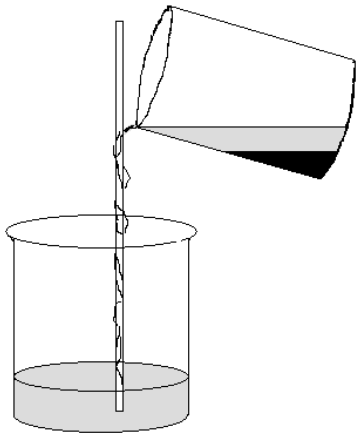
Observations:_____

_____

The heat that you applied has decomposed copper(II)hydroxide, which has "fallen apart" giving copper(II)oxide and water. Write a balanced equation for this decomposition reaction:

_____

The copper(II)oxide is wet with not just water, but also contains excess sodium hydroxide and sodium nitrate from Part 1 and 2. We need to wash the precipitate to separate the copper(II)oxide from the soluble ionic compounds. A filtration would be best but takes time and resources. Repeated decanting can often be employed as a cheap and fast alternative



b) Pour off as much liquid as you can without losing any solid (decanting).
c) Add another 100 mL water to the beaker to wash the precipitate. Stir gently, let settle for 5 minutes and decant again. The solid substance remaining in the beaker is copper(II)oxide.

Part 4 Converting copper(II)oxide to copper(II)chloride.

Measure out about 100 mL of a hydrochloric acid solution, 3 M HCl. (Caution, strong acid.) Pour this acid into your beaker containing the copper(II)oxide. Stir gently until the solution is completely clear.

Observations:_____

The copper(II)oxide has "dissolved" meaning reacted with the hydrochloric acid in an acid base type reaction to produce a salt and water (*metal oxides can be viewed as anhydrous metal hydroxides*). Write a balanced equation for this reaction:

_____

Part 5 Converting copper(II)chloride to copper

a) Measure out about 6 g of zinc metal. Add the zinc to the solution in the beaker and cover it with a watch glass. Touch the beaker to see if the temperature has changed.

Result: _____

Describe the action in the beaker: _____

_____

If the reaction is not occurring rapidly enough add a little more hydrochloric acid. Write a balanced chemical equation for the reaction between zinc and copper(II)chloride.

_____

Cu^{2+}(aq) is being __oxidized/reduced___while zinc metal is being _oxidized/reduced___.

The solution should be losing its colour and excess zinc should have been consumed by the acid leaving the reddish brown copper sponge (very fine copper dust).

The gas that is evolving is hydrogen gas from yet another redox reaction taking place in your beaker simultaneously. Write a balanced equation for the reaction of zinc with hydrochloric acid:

_____

Part 6 Recovery of the copper

a) Decant the clear liquid leaving behind as much of the copper powder as possible. Add about 50 mL of distilled water and wash the precipitate. Allow the copper to settle and decant again. Repeat the washing procedure once more to remove most of the excess acid and salts such as zinc chloride.
b) Determine the weight of a dry porcelain evaporating dish to two decimal places:

   Mass empty evaporating dish:_____

   Transfer all of the copper to this evaporating dish using a rubber policeman and water from a squirt bottle.

c) Place the evaporating dish **on** a glass beaker that holds about 50 mL of boiling water. This steam bath will dry your copper quite quickly. As the contents of the evaporating dish settles draw off as much water as possible using a Pasteur pipette.
   As the copper begins to dry use the glass rod to grind it. Continue heating and grinding until the copper no longer sticks to the rod and is a fine loose powder.
   Remove the dish from the heat and let it cool. Determine the weight of the dish and the product. **Never weigh a hot object!**

   Mass of dish and product:_____

   Mass of copper recovered:_____ % yield of recovery:_____

Part 7 Alternative recovery of copper (demonstration performed by your instructor)

Zinc is a convenient reducing agent in the laboratory but is far too expensive to be used in an industrial process. After metal ores have been roasted (a process that converts metal sulfides to metal oxides by “grilling them in air”) they are often reduced with coal which is an inexpensive readily available and powerful reducing agent (steel manufactory!)

A small sample of copper oxide (donated by one of your) is placed on charcoal and heat is applied with a propane torch until the coal and the sample are above 900 °C. Water from a squirt bottle is used to quench the flames and cool the sample.

Questions: Would each of the following tend to results in: a low recovery, “too high” recovery or no effect on the recovery?

a) Losing some solid during decanting in Part 3 and Part 6

_____

b) Skipping the step “washing of the precipitate” in Part 3 and Part 6

_____

c) Adding more sodium hydroxide solution in Part 2 then was necessarily.

_____

d) Adding not enough sodium hydroxide in Part 2.

_____

e) The clear solution that was decanted in Part 6 still had a greenish/blue tinge.

_____

f) There was still solid zinc mixed in with the copper product after reaction had finished in part 5

_____

g) Not completely drying the copper in Part 6

_____

h) Heating the copper too long and to intense and converting some of the copper to copper oxide.

_____

Lab 7: Thermochemistry

Background

Chemical reactions are accompanied by an energy change. Some reactions give off heat (exothermic), some take up heat (endothermic). Although heat changes do not themselves determine if a reaction is spontaneous or not, useful information can be gained by knowing the amount of heat transferred during the rection.

The **enthalpy change** (ΔH) of a reaction is defined as the heat transfered at constant pressure. Thus, the measure of heat transfer provides the enthalpy of reaction.

To measure heat transfer, it is common practice to use the heat to change the temperature of some known material, often water. Water has a heat capacity of 4.184 J K^{-1} g^{-1}. Any temperature change can be directly related to the heat transfer, except for any heat lost to (or gained from) the surroundings and/or the calorimeter.

LoggerPro

Set up the computer and LabPro with two temperature probes. Start an experiment with an additional calculated column (time-60). Set up the data collection for Parts 1 and 2 for 180s of collection. Set up Part 3 for 1200 s of data collection.

LoggerPro will act as a timer as well as collect the data.

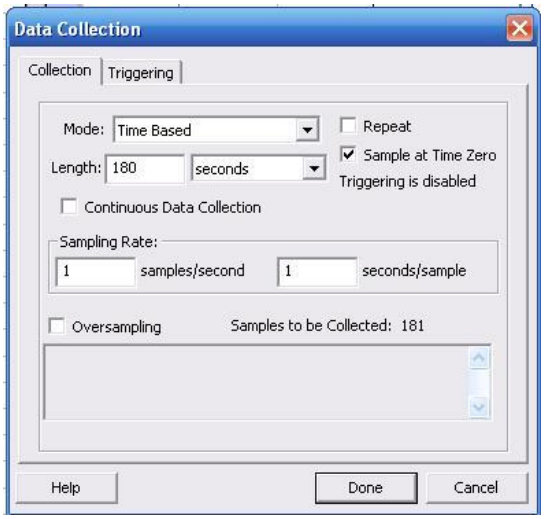
LabPro, the data logger, often crashes if the temperature probes are in the calorimeters when the stirrer is turned **ON** or **OFF**. ***Always*** turn the stirrer on before inserting the temperature probe(s).

Part 1: Calibration of calorimeter

Collect 50 mL of warm DH_2O from the brown bottles in the water bath. Warm means about 40°C.

The calorimeter consists of a glass beaker nested in a Styrofoam cup and a thermometer. To calibrate this calorimeter, measure some temperature changes in water; hot water losing heat (lowering temperature) and cold water gaining heat (rising temperature). Begin with a measured 50.0 mL of room temperature DH_2O in the calorimeter and 50 mL of the warm water in a graduated cylinder. Place a temperature probe in both the calorimeter and the graduated cylinder.

Set up the LoggerPro run with columns for Time, T1, T2, and Count. To add the extra column, in the command line select ***Data***, ***New calculated column***, ***short name*** "timer", ***equation*** "Time"-60 (use ***Variables (Columns)*** to select "Time"). Set up the graph by right clicking on the graph, choose ***Graph options*** then ***x-axis*** choose ***Calculated column***. Set up the data collection with the button . In the Data collection window, select the length to 180s.



.Start the data collection

and watch the time column count from -60. When the timer reaches 0, quickly add the warm water to the calorimeter and place both probes in the calorimeter.

After the run is complete, use ***Experiment, Store latest run*** to save the data as Run 1, and generate a new Latest. To temporarily clear the graph, Enter ***Graph, graph options*** and unselect the Run 1 data from the Y axis. When you start collecting, you will see only the Latest data. Use the same parameters for Part 2 as Part 1.

Part 2: Enthalpy of neutralization

The neutralization of an acid by a base is primarily the reaction of H_3O^+ with OH^-. Place 50 mL of 3 M NaOH in the calorimeter and 50 mL of 3 M nitric acid in a graduated cylinder. Place a temperature probe in both the calorimeter and the graduated cylinder. Start the data collection and watch the time column count from -60. When the timer reaches 0, quickly add the acid to the calorimeter and place both probes in the calorimeter. Collect the data as in Part 1. When finished, start a new Latest by storing the run.

Part 3: Enthalpy of reaction of zinc and copper sulfate

Place **95 mL** of D_{H_2O} in the calorimeter. Precisely (±10 mg) weigh ~5 g **$CuSO_4(H_2O)_5$** in a weigh boat. Add it to the water in the calorimeter and stir until fully dissolved. You now have 100 g solution.

Accurately weigh out ~6.5 g Zn granules in a weigh boat. 0.5 g precision is acceptable for this, as Zn is in large excess.

Save the previous Latest to Run 2, then remove one of the temperature probes. **Set up the new Latest to collect data for 1200s**. Measure the temperature of the copper sulfate solution as in Part 1. At Timer 0 quickly add the Zn, and continue to measure the temperature until the temperature begins to fall. Ensure that the zinc is suspended by swirling the entire calorimeter every once in a while, and check the colour of the solution. All tinges of blue must be gone for the run to be over. You might want to remove the styromoam cup for this part to have the benefit of watching the reaction and ensuring the zinc is stirred sufficiently (not piling up in a corner).

Data analysis and report

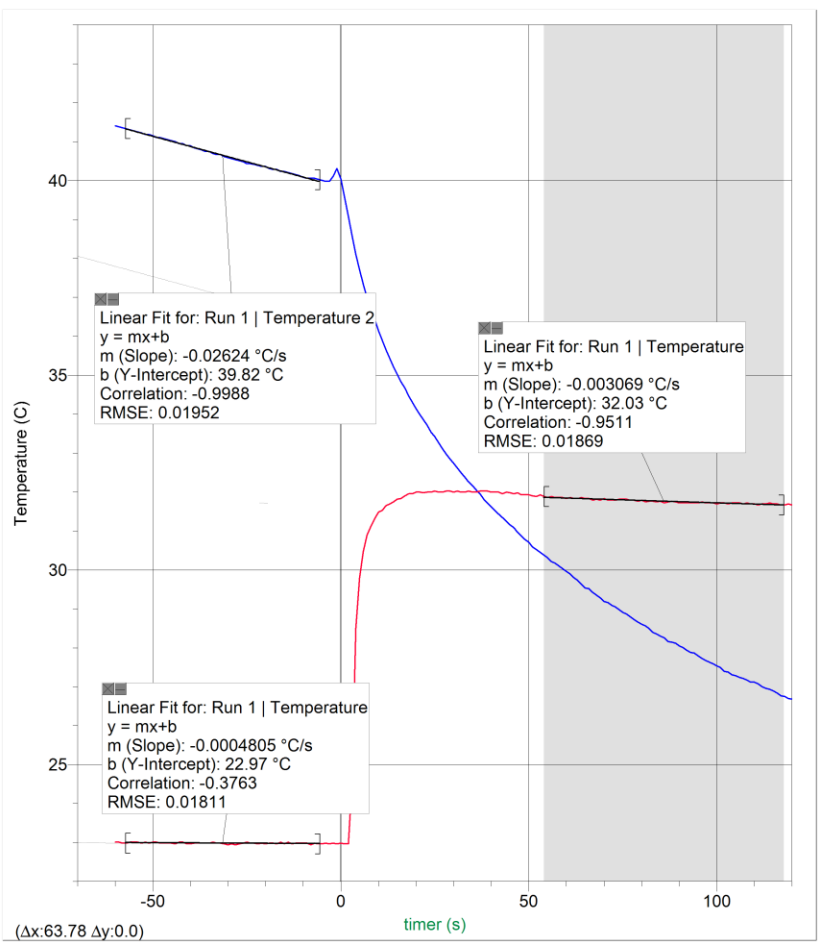
The purpose of the temperature measurements over time is to compensate for the fact that the heat is not necessarily transferred instantaneously throughout the calorimeter, and/or the reaction does not occur instantaneously. By plotting the temperature before and after the addition, the initial limb and the falling limb **can be extrapolated to the time of addition**, which is the estimate for the instantaneous temperature change.

Plot the three exercises using LoggerPro by selecting ***Graph, graph options***. Select the data for each individual run. Use the cursor to select an area in the graph which includes a straight line before timer=0. Select the linear regression tool and select the two temperatures. Select a new area to include the portion when after the temperature has risen then falls in a straight line. Again select the linear regression tool. In order to determine the temperature change at t=0, select the portion of each curve. The intercept in each case is the temperature at time = 0. You should have regressions for all initial and final temperatures.

Copy each graph in turn and import them into an excel spreadsheet from which they can be stored and printed. You need to include the three graphs in your lab report. Complete all calculations in the data table which is attached. You should attach a separate sheet to show your calculations.

Use the heat capacity for water (4.184 J K^{-1} g^{-1}) to determine the heat transfer in the calibration and acid-base reaction cases (Part 1 and 2). For part 3 the heat capacity of the salt metal mixture will not be the same; use an assumed value of 3.97 J K^{-1} g^{-1} for the solution, and 25 J K^{-1} mol^{-1} for the metals (Cu, Zn). This change in heat capacity of the solution accommodates the fact that the solution has a lot of dissolved ions.

Be sure your final answer includes the appropriate number of significant figures and proper units.



Data sheet for Thermochemistry

Parameter	Value	Units
Heat capacity of calorimeter		
Volume of cold water		
Volume of hot water		
Initial temperature of cold water		
Initial temperature of hot water		
Final temperature		
Heat lost by hot water		
Heat gained by cold water		
Heat gained by calorimeter		
Heat capacity of calorimeter		

Heat of neutralization

Volume of contents		
Initial temperature		
Final temperature		
Heat change for contents		
Heat change to calorimeter		
Total heat change		
Molarity of acid/base used		
Moles acid/base used		
Enthalpy of neutralization		

$CuSO_4$ / Zn

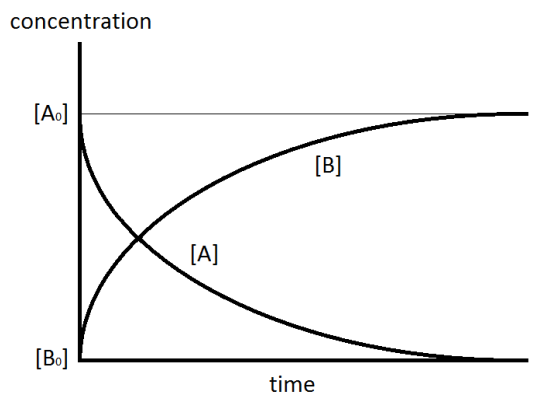
Mass $CuSO_4(H_2O)_5$		
Mass Zn		
Initial temperature		
Final temperature		
Heat change of metal		
Heat change of solution		
Heat change of calorimeter		
Total heat change		
Moles $CuSO_4(H_2O)_5$ used		
Enthalpy of reaction		

Lab 8 Chemical Kinetics - The Bromination of Acetone

Background

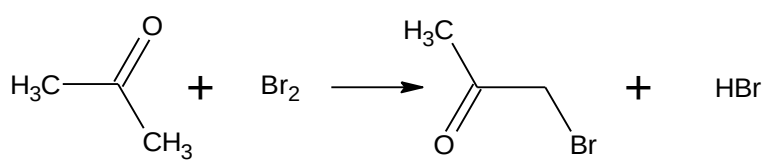
Thermodynamics is concerned with the energy changes associated with a chemical reaction. Chemical Kinetics is concerned with the rates of chemical reactions. The rate of a reaction is the speed at which it proceeds e.g. the change of concentrations of reactants or products as a function of time.

Consider the simple rearrangement if A→B. The rate of the reaction would be $-\frac{d[A]}{dt} = +\frac{d[B]}{dt}$ and the average rate would be $-\frac{\Delta[A]}{\Delta t} = +\frac{\Delta[B]}{\Delta t}$. The graph below illustrates how the concentration of reactant A and product B change over time.



Determining rates of reactions and activation energies is done by first determining the order of reaction for all components, then determining the rate of reaction at different temperatures. Orders express how reactants influence the rate of the reaction. We will get there soon.

We will examine the reaction between acetone (CH_3COCH_3) and bromine (Br_2), which follows the following stoichiometry (acid is also needed to catalyze the reaction):



The rate law will look something like:

$$\textbf{Rate}=k[\text{acetone}]^a\,[\text{bromine}]^b\,[H^+]^c$$

where a, b, and c are the orders of reaction for each component, and initially unknown. K is the rate constant for the reaction and also unknown.

One way of determining orders is to sequentially change concentrations of the participants, and observe the change in rates. For 0-order, the rate is independent of the concentration, so no change in <u>rate</u> will occur; for 1st order, doubling the initial concentration doubles the rate; for 2nd order, doubling the initial concentration quadruples the rate. (Confirm this for yourself using the above rate law equation.)

Once the rate law is known, the rate constant **k** can be determined. k is a natural constant for every reaction and if the temperature is constant the rate constant will be a true constant. Conducting the procedure at more than one temperature will permit the determination of a set of **k** values, which can be used to determine the activation energy.

The equation which governs activation energy E_a is:

$k = A \exp(-E_a/RT)$ where k is the rate constant R is the gas constant, T the Kelvin temperature and

A is the pre-exponential factor. (It is related to the fact that not every collision between reactants that is energetically enough results in a reaction. For some reactions the molecules also have to collide with the right orientation.)

Taking natural logs of both sides, we get:

$$\ln k = \ln A + \left(\frac{-Ea}{R}\right)\left(\frac{1}{T}\right)$$

which is a linear equation of the form y=mx+b, where m is the slope and b the y intercept. For our case, y =ln k, b = ln A, x=1/T and m = $-E_a/R$.

Procedure and data Collection

Bromine is a coloured compound; it's the only coloured compound in our reaction so its disappearance can be monitored by loss of colour. If the initial concentrations of bromine are very low compared to the other participants, then the concentrations of those other participants can be taken as constant for any single case. The average rate can therefore be estimated as the time it takes for the bromine to disappear; i.e. $\Delta[Br_2]/\Delta t = ([Br_2]_f - [Br_2]_i)/(t_f - t_i)$. However, $[Br_2]_f = 0$ and $t_i = 0$, so the rate is:

$$\textbf{Rate} = -[Br_2]_i/t_f$$

Set up a series of six 100 mL conical flasks with volumes of acetone, water and HCl as per the data table and spreadsheet (see below). Label each flask! Sequentially, add the specified amount of bromine solution to the conical flask <u>and start the timer</u>. Mix well and let the flask stand on white paper to see the colour better. You should end up with a total volume of 50 ml for each run as you add the bromine solution. You will find it useful to tick off each component as you add it. Forgetting, or adding too much, of any component will make data interpretation very difficult to impossible. I encourage you to start multiple runs and monitor them simultaneously for the endpoint. Determine the time required to react all of the bromine for each run. Having a conical flask with a finished reaction as a blank to compare the colour of the current reaction will be very helpful.

Four of the reactions are done at room temperature, one held cold (≥10°C) and one warm (~40°C). Your instructor will monitor the cold and hot bath and maintain the temperature as constant as possible. The precise temperatures for hot and cold baths will be announced in the lab.

Write all raw data in the data table/spreadsheet!

Data analysis

Fill in the data tables provided; determine the orders of reaction (a, b and c) and the activation energy. To determine the activation energy, plot *ln k* vs 1/T (the inverse Kelvin Temperature) for run #3, #5 and #6. Only those runs can be used because the temperature was the only variable whilst all concentrations of reactants were kept constant.

Submit your data sheet with all the relevant fields filled in and attach your graph. Ensure that your graph has a meaningful title and label the axis properly. Give the orders a,b and c (one significant figure!) as well as the slope of your plot and the activation energy. Include proper units.

Data sheet for Bromination of acetone

#	T °C	Vol. of acetone	Vol. of HCl	Vol. of Br_2	Vol. of H_2O	ΔTime in s	$[C_3H_6O]$ in mol/L	[HCl] in mol/L	$[Br_2]$ mol/L	Rate = $-[Br_2]_i/\Delta time$
1		10 mL	10 mL	10 mL	20 mL					
2		20 mL	10 mL	10 mL	10 mL					
3		10 mL	20 mL	10 mL	10 mL					
4		10 mL	10 mL	20 mL	10 mL					
5	Cold	10 mL	20 mL	10 mL	10 mL					
6	Hot	10 mL	20 mL	10 mL	10 mL					

Continue horizontally with:

#	$Rate_n/Rate_1$ (1 sig fig)	$k = \frac{-Rate}{[acetone]^a[Br_2]^b[HCl]^c}$	ln k	1/T in1/K
1				
2				
3				
4				
5				
6				

Orders: a=_____b=_____c=_____

$\ln k = \ln A + (-E_a/R)\ 1/T$

slope = _____

E_a = _____

Lab 9: Gravimetric Analysis of Chloride

Note: This procedure is to be done by ***all*** students, although teamwork is encouraged. **Each** student will submit **two** results.

Background

Perhaps the simplest way to determine the amount of a specific target compound or ion is to prepare an insoluble compound which includes the target and determine the mass of the insoluble compound, then the mass of the target. If the target is in solution, the reaction with a reagent, which will precipitate the target, can be straightforward.

Chloride (Cl^-) reacts with silver ion (Ag^+) to form insoluble silver chloride (AgCl). The silver chloride precipitates out of solution and can be filtered, dried, and the mass determined.

The concentrations of Ag^+ and Cl^- in solution are determined by the solubility product of AgCl:

$$K_{sp} = [Ag^+][Cl^-] = 1.8 \times 10^{-10}.$$

Clearly, if Ag^+ is in significant excess, the concentration of Cl^- remaining in solution will be very low, ensuring that virtually all of the target will be collected as precipitate.

AgCl is first precipitated as a colloid, which consists of very small solid particles; small enough to pass through a filter. Colloid particles are held apart by charges on the particle surface, so removal of the charges can result in the coagulation of the particles until they are large enough to be retained by the filter. This is done by adding electrolyte to the mixture and gently heating.

One problem with this method is that, during colloid formation and coagulation, other compounds, such as $AgNO_3$, may be captured when the precipitate forms.

Most silver compounds are light sensitive, with the Ag^+ being reduced to Ag metal, which appears purple or ultimately black.

Procedure

You may work in pairs, but ***each*** student will do ***two*** analyses***.***

Precisely weigh out a sample of approximately 0.4 g of the unknown into a 400 mL beaker. Do not cross contaminate your unknown samples. Use a dry and clean spatula for each measurement. To the beaker add 200 mL DH_2O and 5 drops of 6M nitric acid (HNO_3). Slowly and with stirring, add 5%(w/v) $AgNO_3$ until the precipitate begins to coagulate, then add 5 mL more $AgNO_3$ solution, also slowly. Heat the mixtures on hotplates until the mixture is *almost* boiling, and simmer for a few minutes. The solution should clear, with a slightly purple precipitate on the bottom.

Whenever the solution clears or settles, add a few drops of the $AgNO_3$ solution to test that no more precipitate forms. If more precipitate appears, add $AgNO_3$ until there is no more precipitation.

Place the mixtures in a dark cupboard for 30 minutes. Repeat the procedure for your second trial while the precipitate of your first trial is aging. By the time your 2nd trial is ready to go in the dark cupboard you should be able to filter your first trial.

Label a filter paper and determine its precise masses. Prepare this paper following the instructor's demonstration. Decant as much of the supernatant from the beaker through the filter. Wash the precipitate with 0.03M HNO_3 and decant into the filter. Add all of the precipitate to the filters. Try to use small quantities of 0.03 M HNO_3 to flush most of the precipitate out of the beaker and onto the filter. The rubber policemen should only be employed as a last resort if AgCl(s) is stuck to the beaker (it also gets stuck on the rubber). Wash the beakers and remaining precipitate into the filters with a minimum amount of 0.03M HNO_3. Wash each filter with about 15 mL DH_2O.

Spread the filter paper out on a watch glasses and place in the oven. Repeat for your second trial. The filters will be removed the next morning and placed in the fume hood. You will weigh the precipitate and complete the calculations during the next day (or as arranged with your instructor).

Data sheet for the determination of chloride

Name: _____ Date: _____

Unknown #: _____

Filter #	**Mass of unknown (g)**	**Initial mass of filter (g)**	**Final mass of filter (g)**	**Mass of AgCl (g)**	**% of Cl^- in unknown**

Lab 10: Atomic Absorption Spectroscopy - Detecting Copper Ions in Municipal Drinking Water

Introduction

You will have already learned in this course that when electrons in gaseous atoms of elements are excited from an orbital of lower energy (the ground state) to an orbital of higher energy (excited state) they absorb a very specific quantum of energy, one that corresponds precisely to the difference in energy between the ground state and the excited state. After a very short time (about 10^{-8} s) the electron usually relaxes back to the ground state. Since energy must be conserved this process is accompanied by the electron emitting radiation of the very same energy, corresponding to the differences in energy between the excited state and the ground state. The figure bellow (Petrucci) represents a selection of the possible transition of the electrons in a hydrogen atom from an excited state to a lower energy state. For the transitions that temporarily "deposit" the electron in the 2nd principle shell (e.g. in an orbital with the principle quantum number n=2) the energy radiated lies in the visible region of the electromagnetic spectrum (Balmer Series). Use Planks equation and check for yourself that the transitions from n=5 to n=2, n=4 to n=2 and n=3 to n=2 produce photons of visible light.

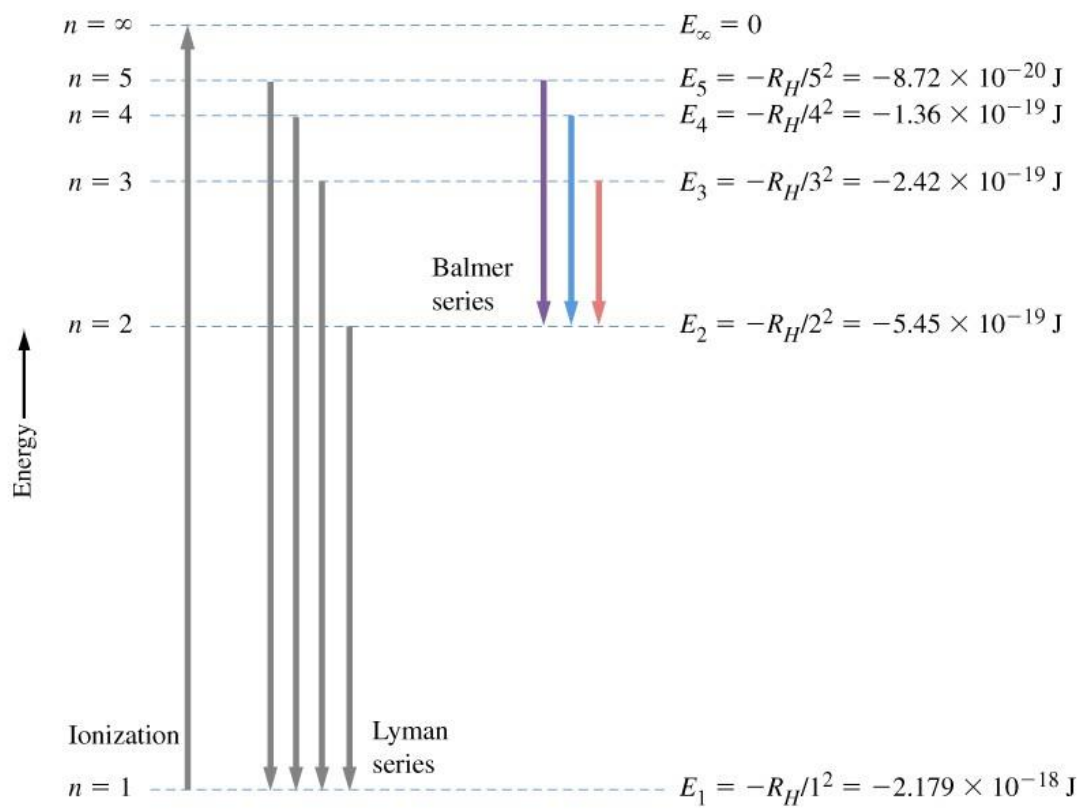


Figure 1: Energy level diagram for the hydrogen atom. If the electron acquires 2.179 x 10^{-18} J of energy it moves to the orbital n=∞ and ionization of the atom occurs H(g) + I_1 →H^+(g) + e^-

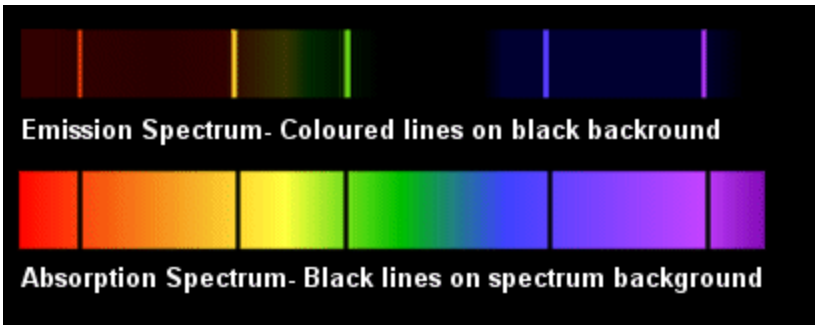


Figure 2: Emission and absorption spectrum of Hydrogen (4 college.co.uk)

In an emission spectrum a series of lines are observed at well defined wavelengths which collectively is called an emission line spectrum. Each element has a unique line spectrum and so it can be used in the qualitative determination of elements in samples. Conversely, if a gaseous sample of the element is exposed to electromagnetic radiation it absorbs light of the same wavelengths which are observed to be emitted in the emission spectrum of the element. Because these absorptions obey Beer's law it is possible to use them in the quantitative determination of an element.

Since 1955 the technique of atomic absorption spectroscopy has made a major contribution to the revolution that has occurred in the area of trace metal analysis. More than fifty different elements can be analysed using this technique. Many of these elements can be measured down to very low concentrations, typically to a few parts per million or even lower. Because of the inherent selectivity of the method it is possible to use simple sample preparations and to carry out the measurement in a matter of seconds. In this way, the overall time required for a single analysis is extremely short. When this time saving is coupled with the method's great sensitivity it is easy to understand why all laboratories engaged in routine or semi-routine elemental analysis find this technique attractive. For each element there is an associated hollow cathode lamp which, when it is lit, emits radiation characteristic of that one element. A liquid sample is aspirated (sucked) into a flame with a very stable temperature, typically between 2000 K and 3000 K. The high temperature causes the sample to be atomised (broken into atoms) in the flame. The flame essentially replaces the cuvette used in conventional spectrophotometry (e.g. the cuvette you used to hold the nickel solution in your spec 20 experiment). The path length of the flame is typically 6 or 10 cm. The radiation from the lamp passes through the flame as a collimated (parallel) beam and at the monochromator a particular wavelength is selected and allowed to pass through to the photo detector. The wavelength selected corresponds to a line (usually the most intense) in the absorption spectrum of that element. In the case of copper the line chosen is at 324.7 nm. The resulting signal is amplified and fed to a recorder. A highly simplified drawing of the essential

components of a typical atomic absorption spectrophotometer is shown below

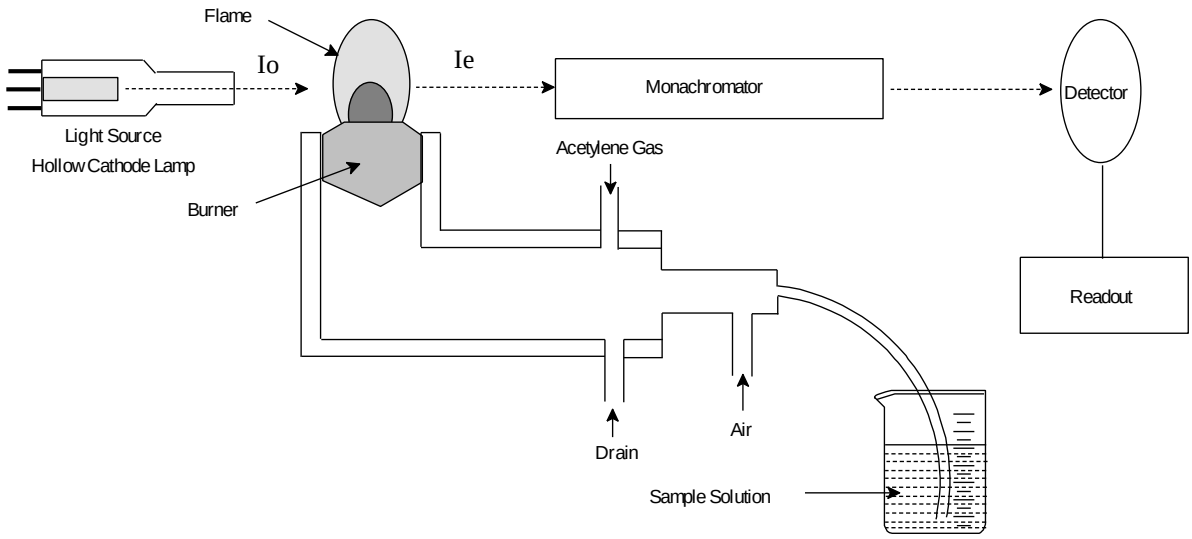


Figure 3: Schematic Diagram of an Atomic Absorption Spectrophotometer. From Camosun Chem 120 Lab manual

In order to carry out an analysis some deionized water is aspirated into the flame by means of a nebulizer. For accurate measurements it is essential that the flame is at a steady temperature and well defined so that the path length of the light beam is constant. Under these conditions the meter is adjusted to zero absorbance. The blank is removed and the calibration standards and the test sample are aspirated into the flame in the same manner. The element of interest present in the standards and in the sample will absorb some of the radiation from the lamp and reduce the intensity of the light beam arriving at the detector. This decrease in radiation intensity follows Beer's Law which states that the absorbance is directly proportional to the concentration of the atoms in the flame. However, the excited atoms must eventually lose their excess energy and one of the many processes of doing so involves an excited atom emitting light of the same wavelength as that used to excite it in the first place. In the unlikely event that all the excited atoms lost their excess energy in this manner, an absorption of radiation would still be detected and recorded. The reason for this apparent contradiction lies in the fact that the emitted light is dispersed uniformly in all directions and hence only a small fraction of the total emission enters the detection system. As the number of such emitted photons of light is also directly proportional to the concentration of the atoms in the flame Beer's Law is still obeyed.

Experimental Procedure

1. Prepare a standard solution of copper by weighing out (to 3 decimal places on an analytical balance) between 125 and 155 mg of copper metal and transferring it quantitatively to a 100-mL volumetric flask. Take the flask to the **fumehood** and add 5 mL of 6 mol/L HNO_3.

Caution! The nitric acid vigorously reacts with the copper and evolves toxic brown fumes.

2. Once the metal has **fully dissolved** (check thoroughly for any undissolved solid!) make the solution up to the mark with deionized water at the fumehood. Stopper and mix thoroughly.

This standard solution must be diluted by a factor of 100 in two steps as follows.

3. Rinse a pipette with a small amount of your copper standard solution. Pipette a 10-mL aliquot of the standard solution into a clean 100-mL volumetric flask, dilute to the mark with deionized water, stopper and mix thoroughly.

4. Rinse the 10 mL pipette with the diluted solution obtained above. Pipette a 10-mL aliquot of the diluted standard solution prepared in step 3. Transfer this to a clean 100-mL volumetric flask, dilute to the mark with deionized water, stopper and mix thoroughly. **Label your volumetric flasks so you can distinguish between them unambiguously.**

5. Pipette 1, 2, 5 and 10 mL of the 100-fold diluted standard copper solution prepared in step 4 into four labelled 50-mL volumetric flasks. Carefully make each flask up to the mark with deionized water. Stopper and shake well.

6. Transfer your unknown sample (e.g. drinking water you brought from home or a sample of the college's water from the tap) into a labelled carefully rinsed 100 mL volumetric flask.

With the instructor present measure the absorbance of each standard and the unknown using the AAS instrument.

Data analysis and report: Generate a graph of absorbance vs concentration of your four copper standards in mg/L (ppm). Include zero! Find the trend line and use the equation to calculate the concentration of copper in your water sample. Label your axis and give your graph a meaningful title.